

Measuring the *Depth* of LLM Unlearning via Activation Patching



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Can an LLM really forget?

LLM unlearning removes specific knowledge (a forget set) from a model while keeping its performance.



What is the profession of Hsiao Yun-Hwa's father?



Before

The father of Hsiao Yun-Hwa is a **civil engineer**.

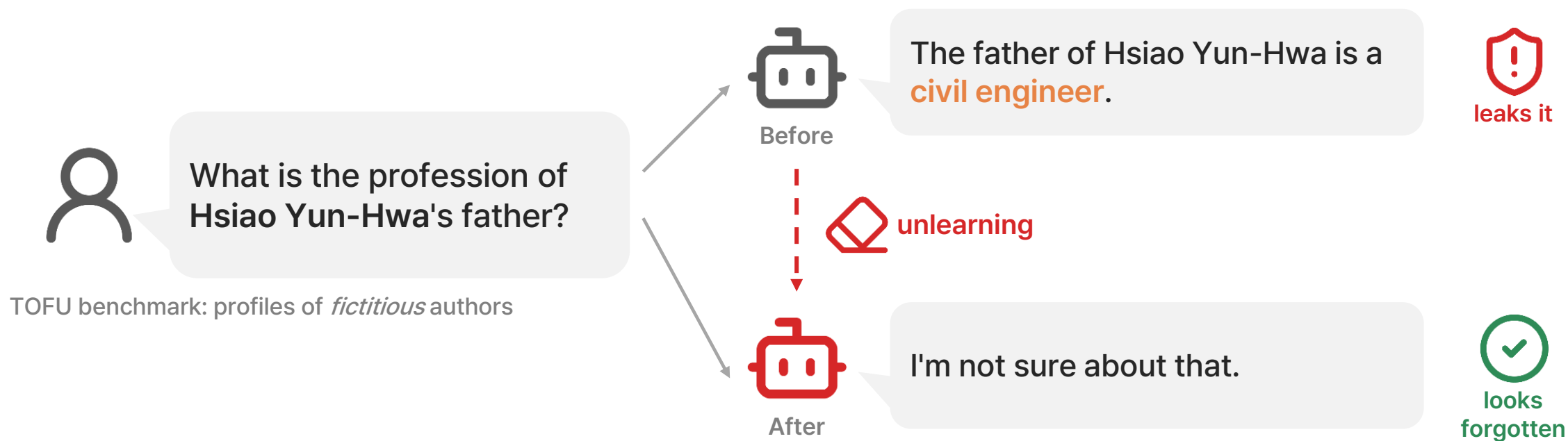


leaks it

TOFU benchmark: profiles of *fictitious* authors

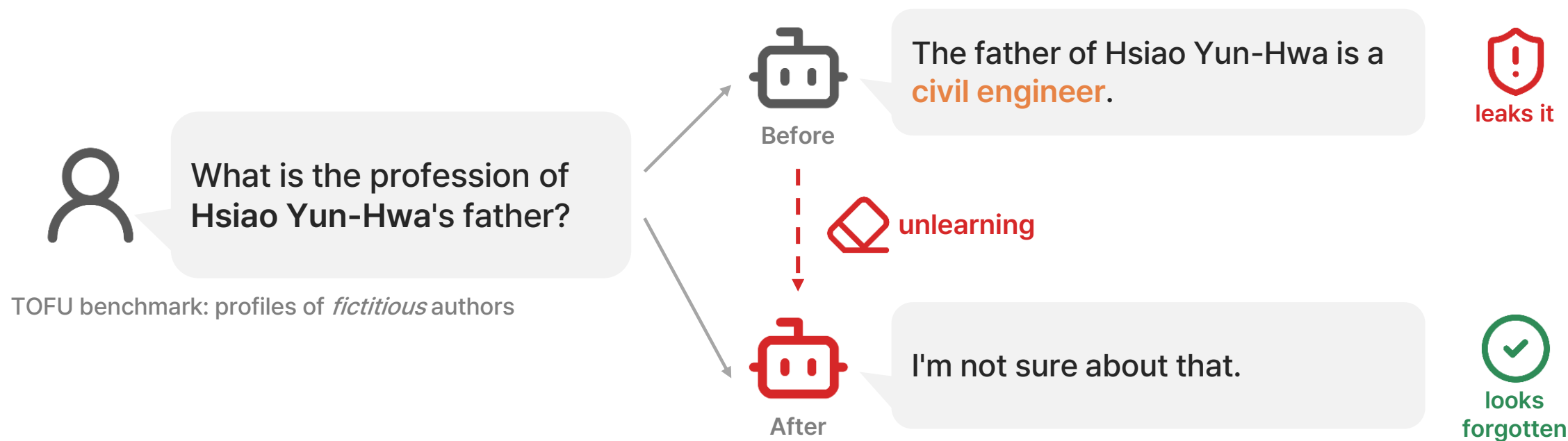
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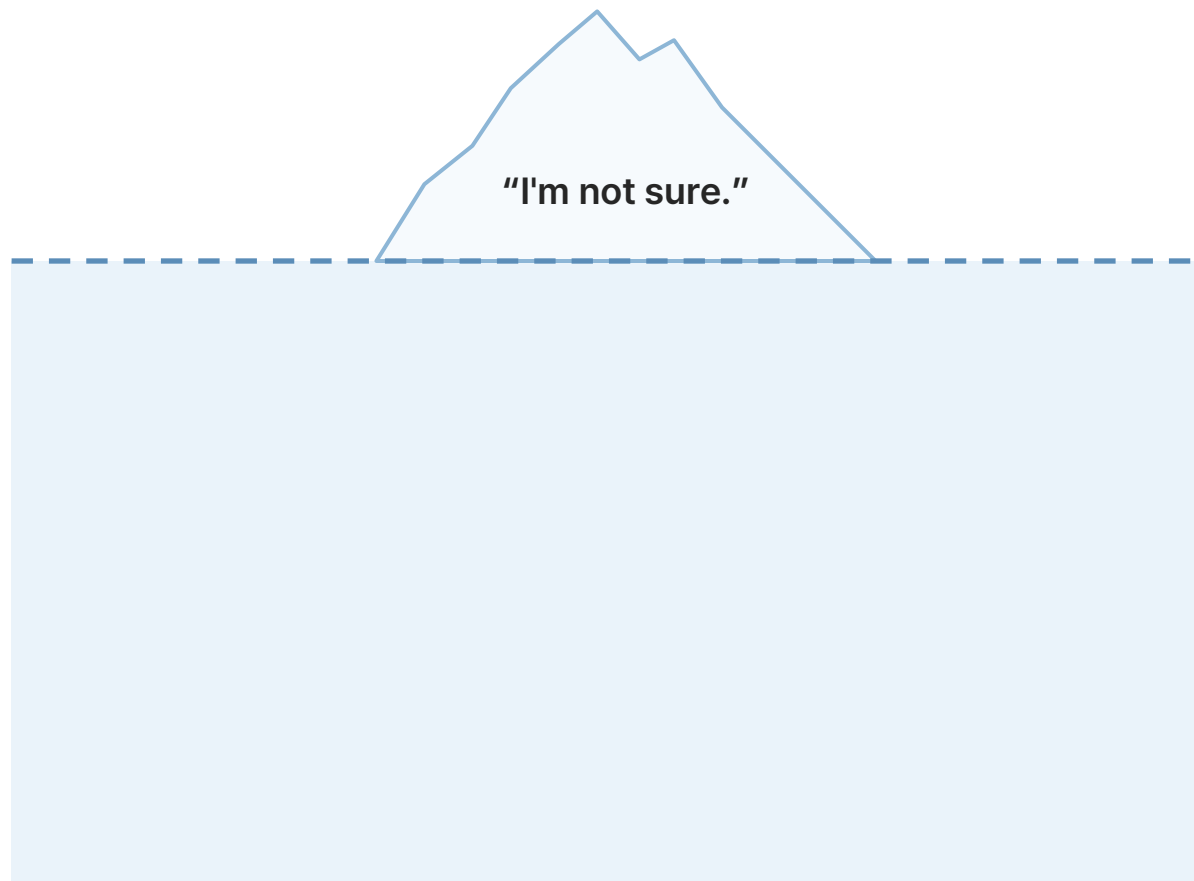
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...but did it really forget, or just learn to stay quiet?

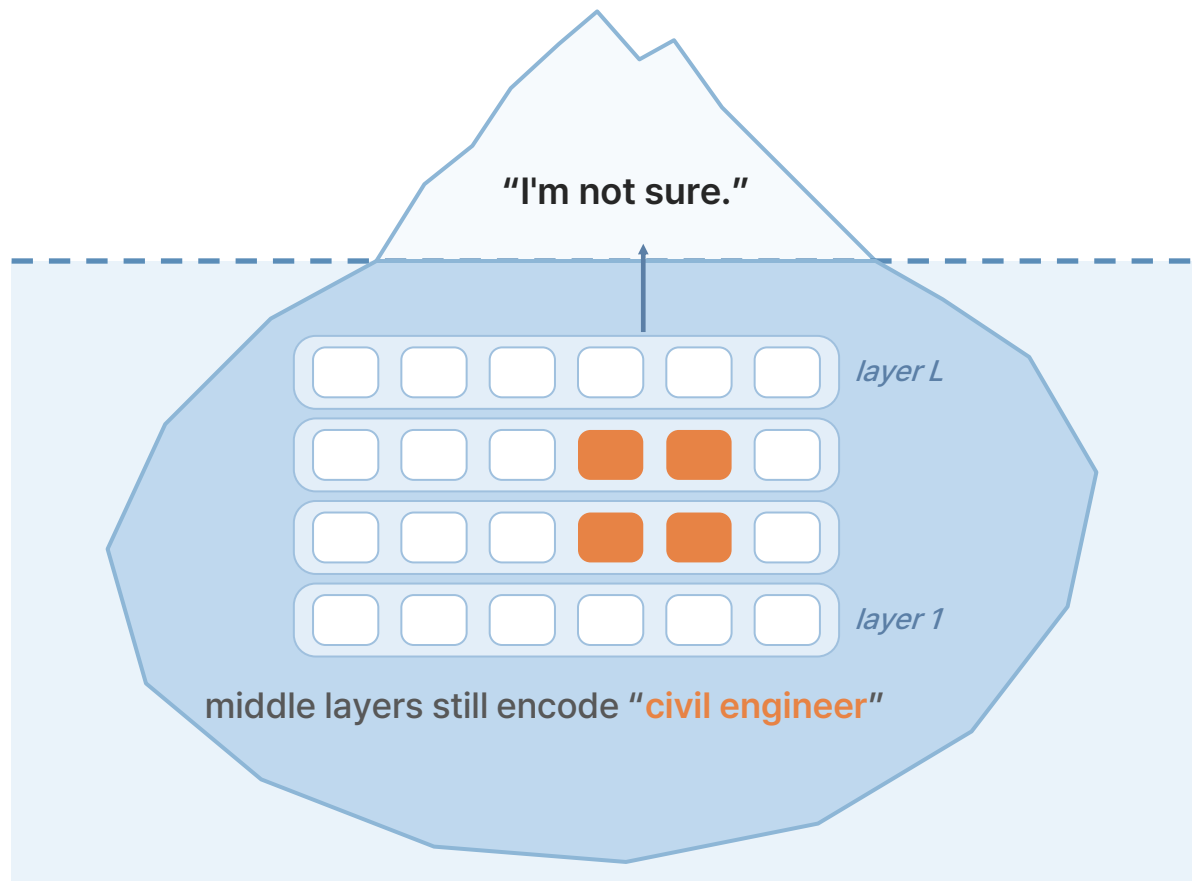
What the model **says** $\neq$ what it still **knows**



What it says

Output-level metrics: ROUGE · Truth Ratio · MIA ...

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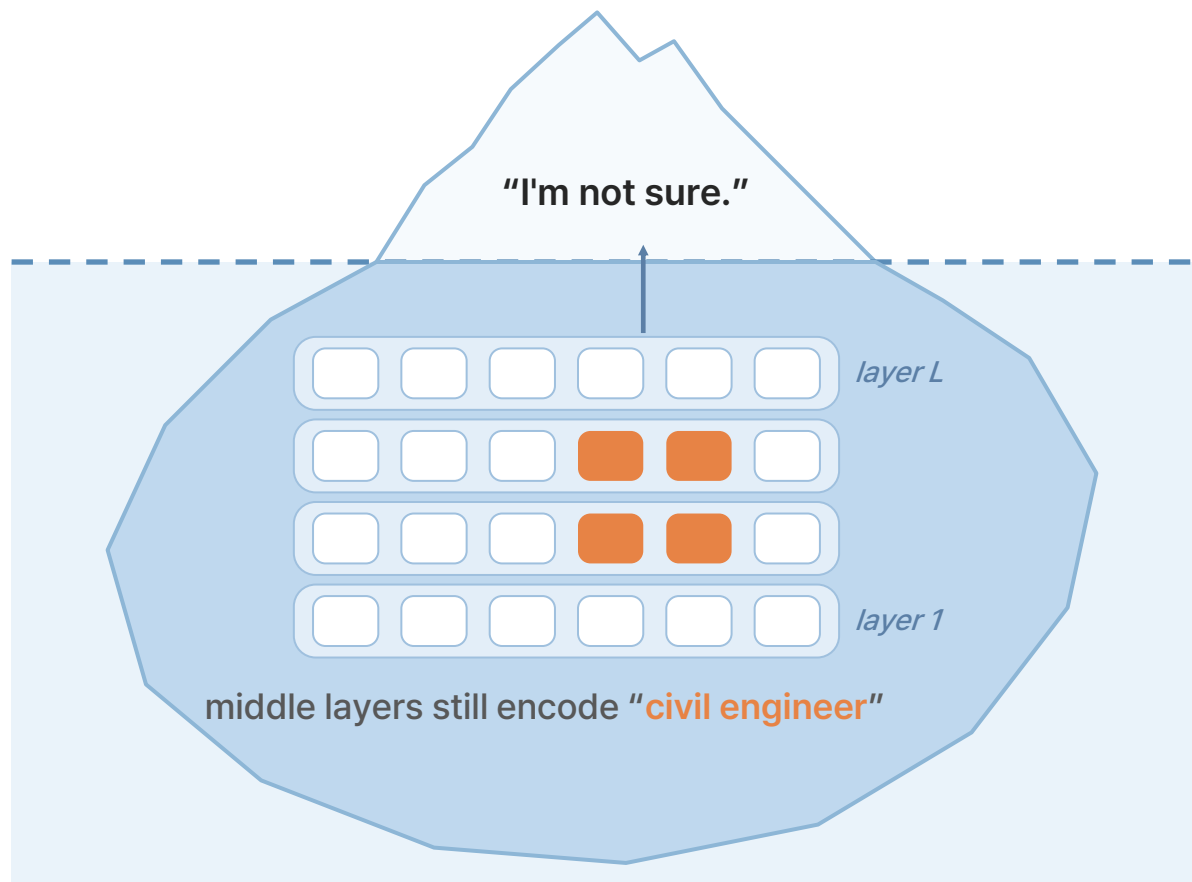
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"Forgotten" facts come back with

- **a little fine-tuning** (Fan et al., 2025)
- **activation steering** (Seyitoğlu et al., 2024; Jang et al., 2026)

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We have to look below the surface, inside the model

Prior white-box analyses: **diagnostic**, not a metric

Work	How it looks inside	Training-free	Causal	Dataset-invariant	Score
Xu et al. (2026)	CKA + Fisher diagnostics	✓	✗	✓	✗
Lynch et al. (2024)	trained probes	✗	✗	✓	✗
Guo et al. (2025)	causal tracing + probes	✗	△	✗	✗
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△ causal localization, but observational evaluation

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UDS (Ours)	two-stage activation patching	✓	✓	✓	✓

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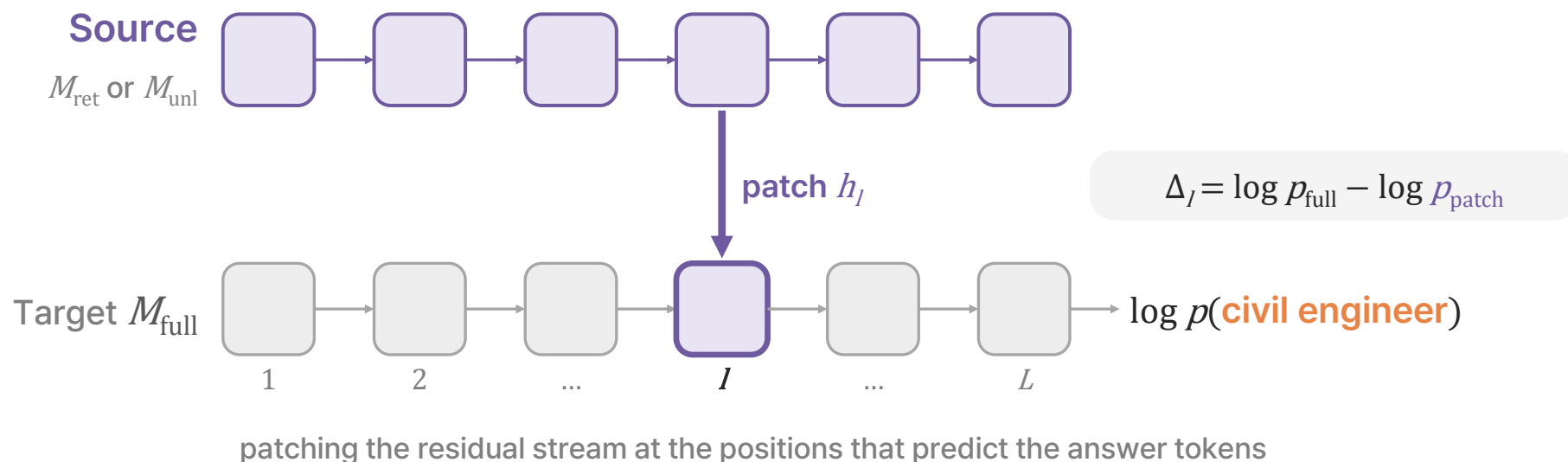
Key idea: can the full model still read the fact?

M_{full} trained on all data
knows the fact

M_{ret} trained without forget set
gold standard

M_{unl} after unlearning
forgot it?

Input (teacher forcing): "...Hsiao Yun-Hwa is a **civil engineer**"



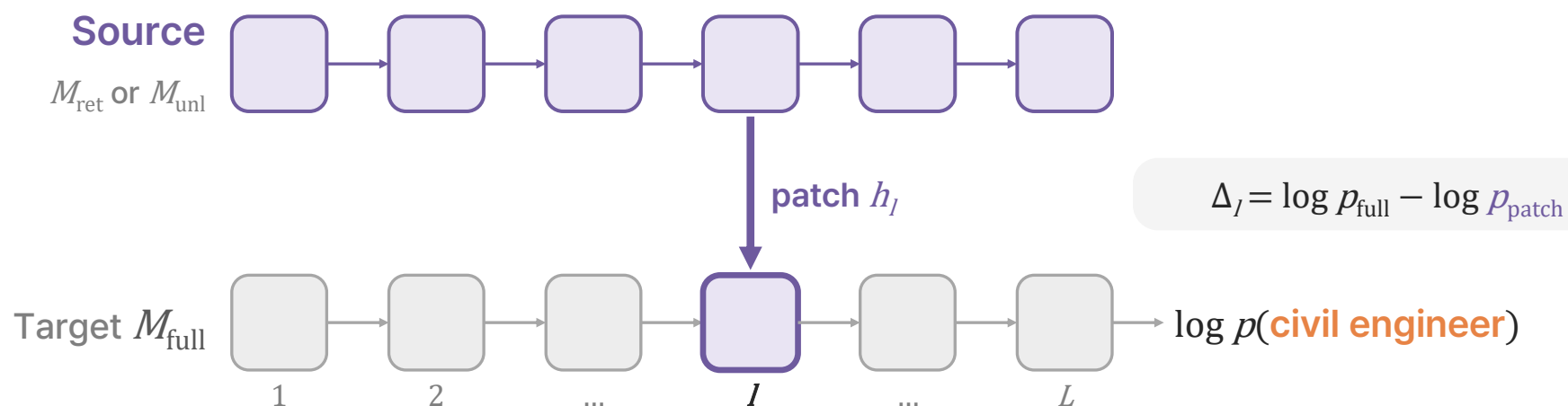
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patching the residual stream at the positions that predict the answer tokens

Source still holds it
→ M_{full} decodes it:
small drop

Source lost it
→ prediction collapses:
large drop

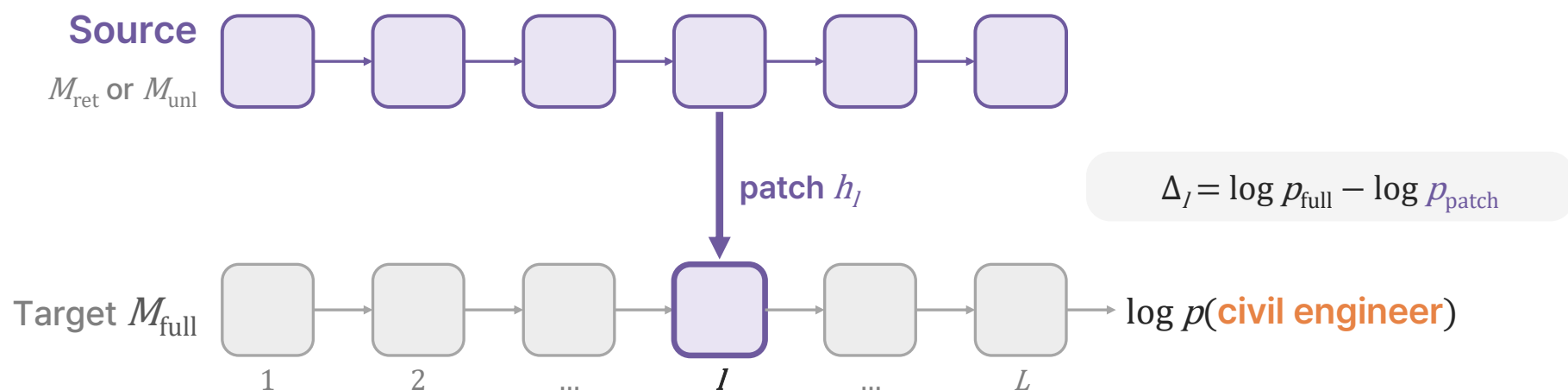
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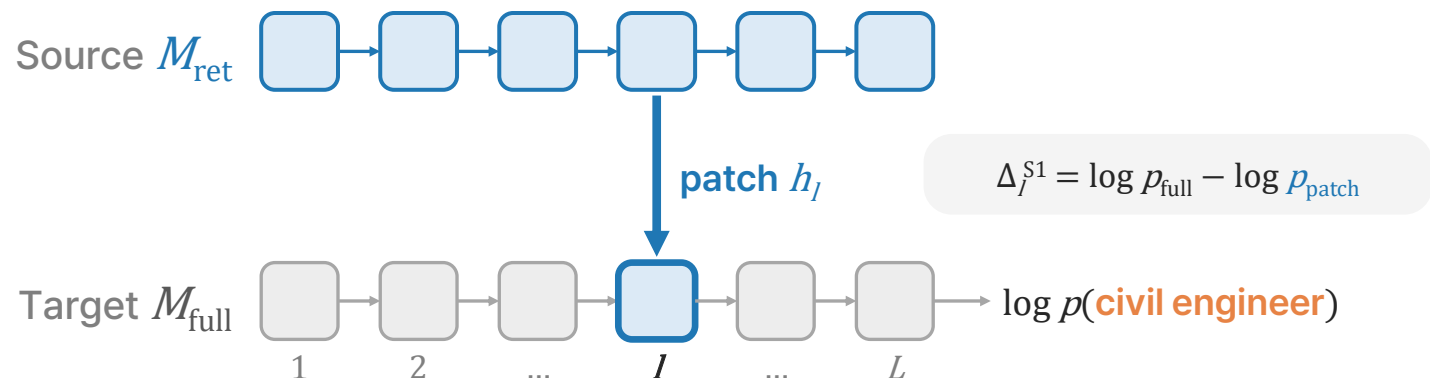
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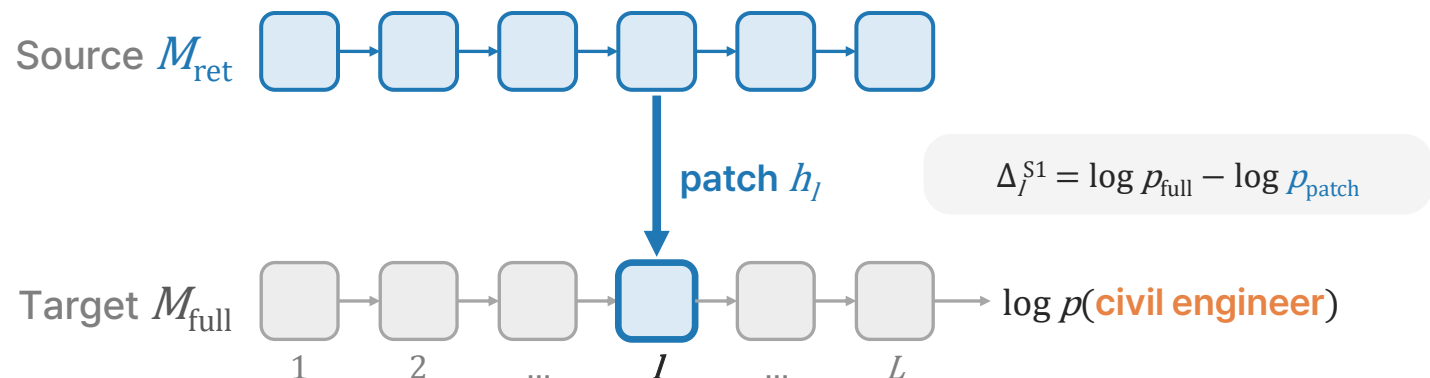
The full model acts as a reader

Stage 1 · Baseline: where is the fact encoded?



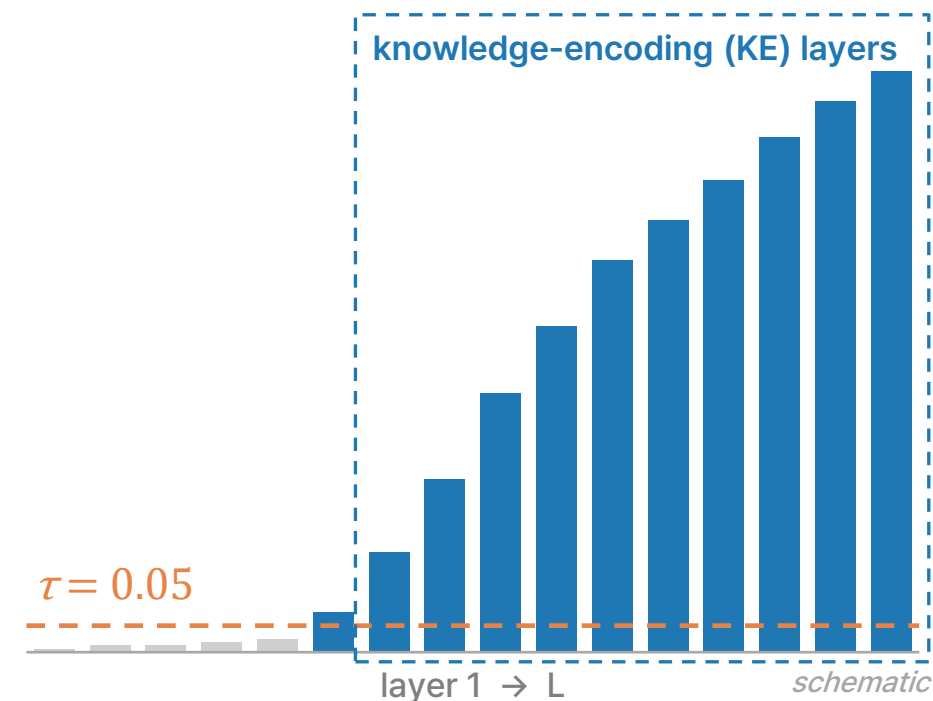
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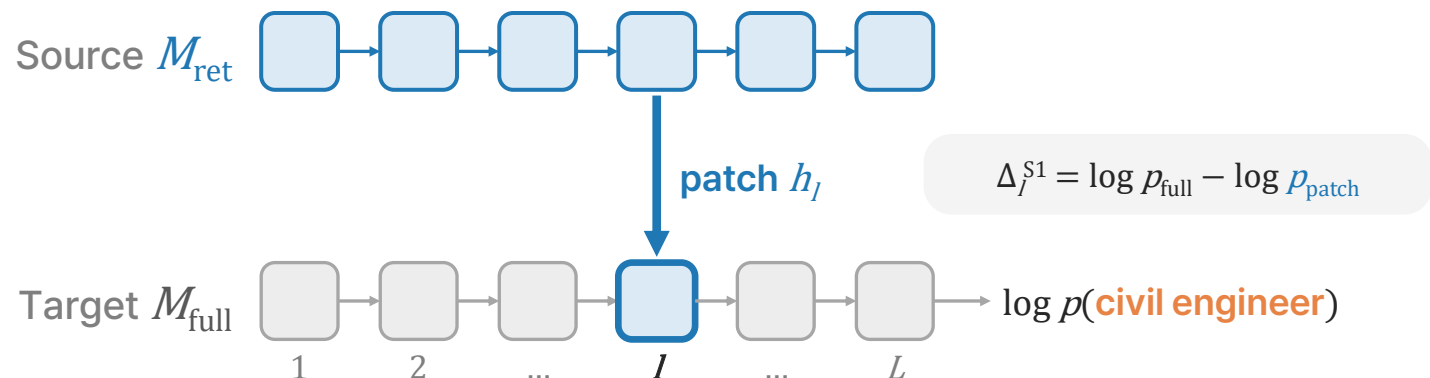


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Δ_I^{S1} drop when patching M_{ret} at each layer

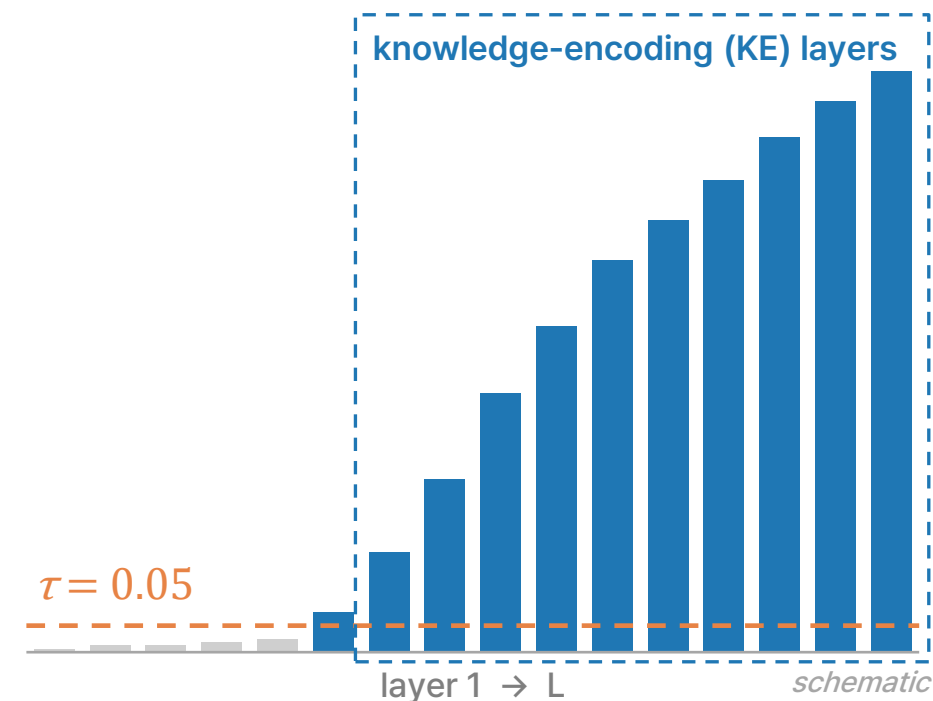


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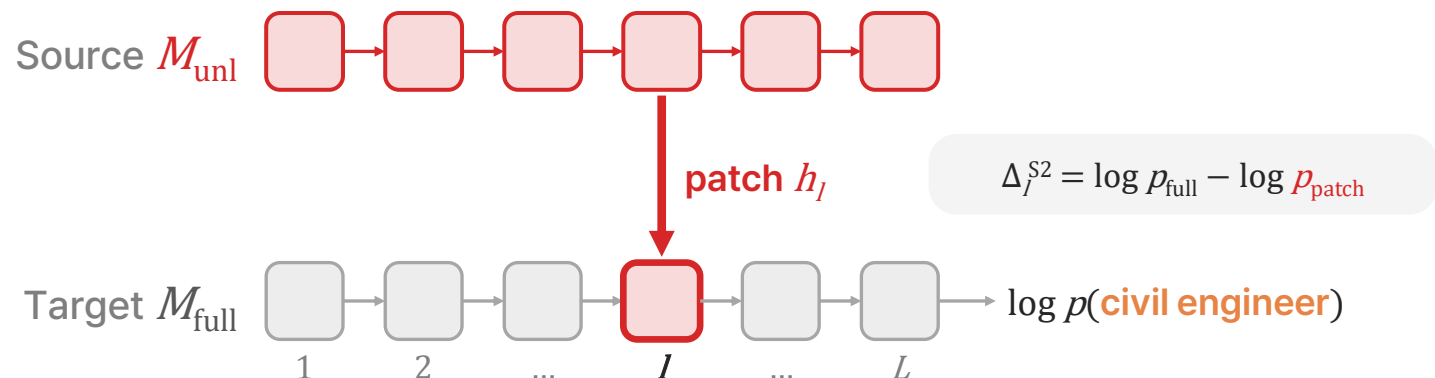
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Stage 1 = what full erasure looks like

Stage 2 · Quantification: is it still there?

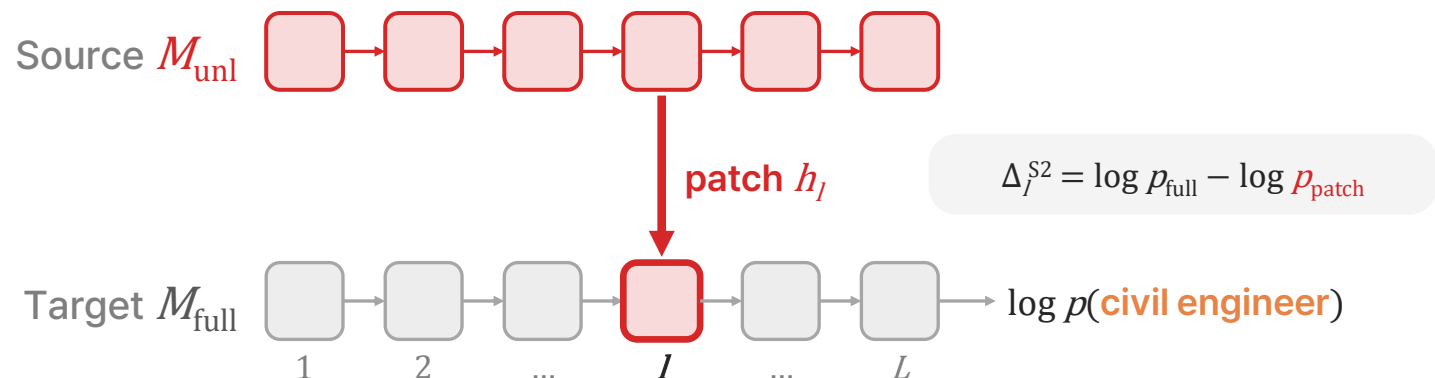


- Same procedure, source = M_{unl}
- Truly erased $\rightarrow$ hurts M_{full} as much as M_{ret} did
- Still inside $\rightarrow M_{\text{full}}$ decodes it: no drop

schematic

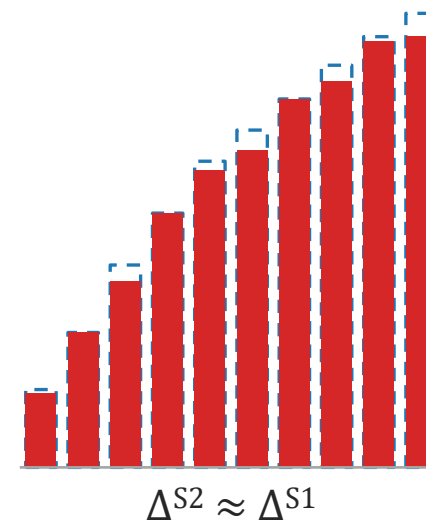
Δ^{S1} Stage 1 (KE layers) Δ^{S2} patching M_{unl}

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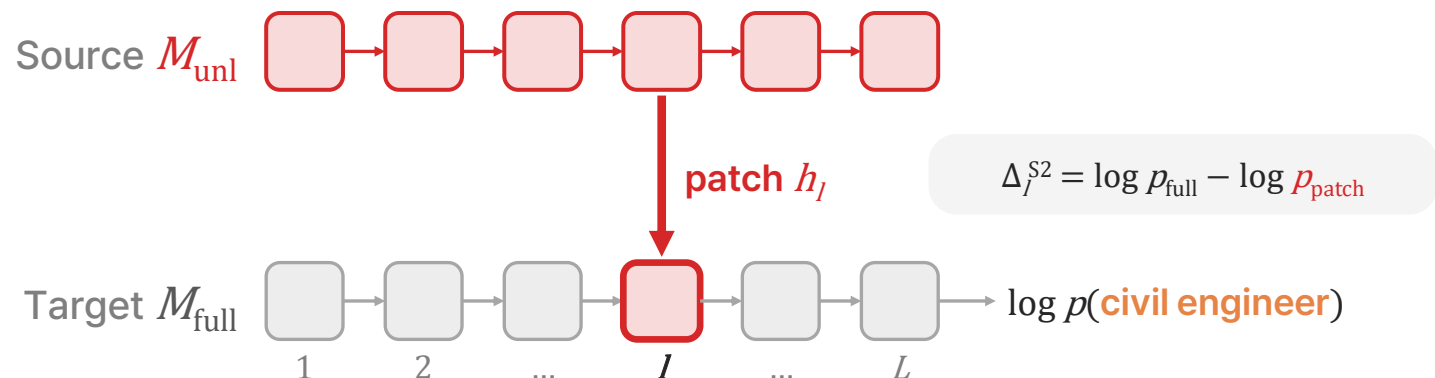
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Erased

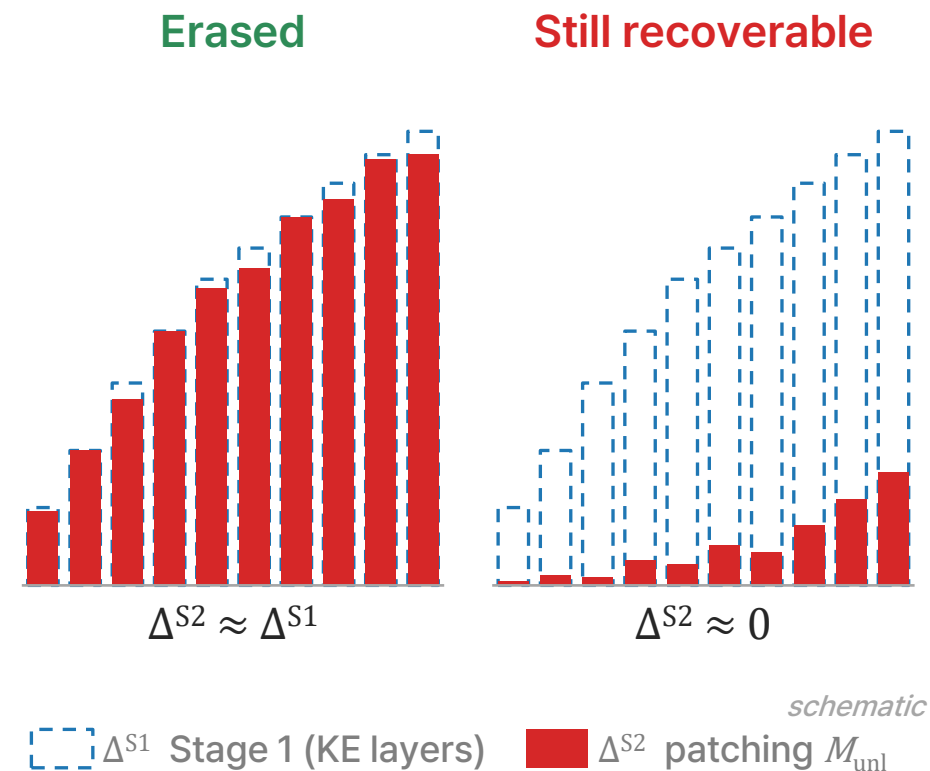


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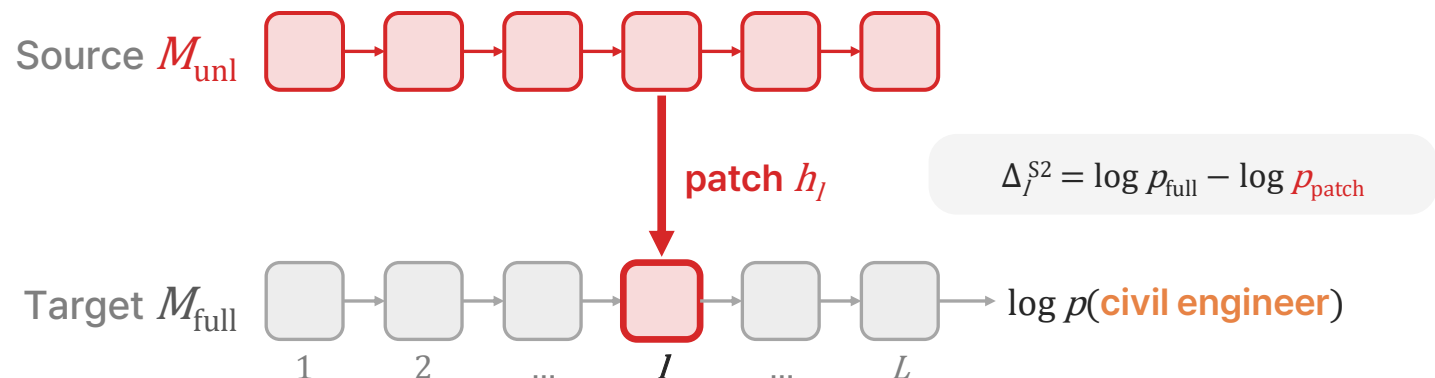
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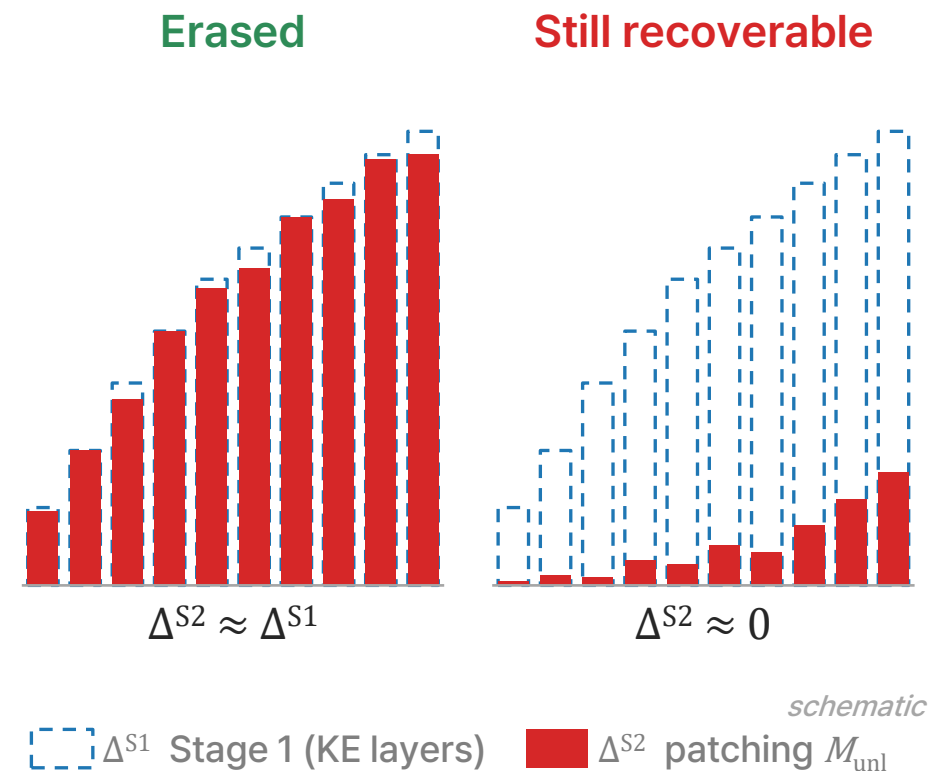
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Stage 2 · Quantification: is it still there?



- Same procedure, source = M_{unl}
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Does M_{unl} hurt M_{full} as much as M_{ret} ?

From layers to one score: UDS $\in [0, 1]$

1. Layer Erasure Ratio

$$\text{LER}_{i,l} = \text{clip}\left(\frac{\Delta_{i,l}^{S2}}{\Delta_{i,l}^{S1}}, 0, 1\right)$$

how much of the retain-level drop the unlearned model reproduces at layer l

2. Unlearning Depth Score

$$\text{UDS}_i = \frac{\sum_{l \in \text{KE}_i} \Delta_{i,l}^{S1} \cdot \text{LER}_{i,l}}{\sum_{l \in \text{KE}_i} \Delta_{i,l}^{S1}}$$

weighted by how much knowledge each layer holds; model score = mean over examples

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Training-free · causal · per example & per layer

Meta-evaluation: is UDS a reliable metric?

Faithfulness

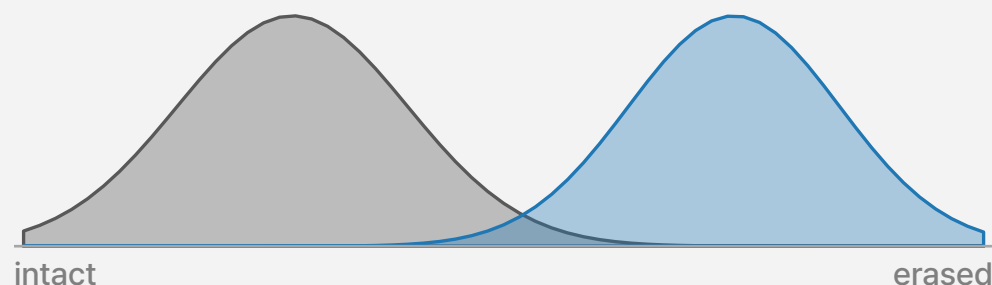
Does it tell apart models that know vs. don't know?

P-pool

30 models trained with D_f

N-pool

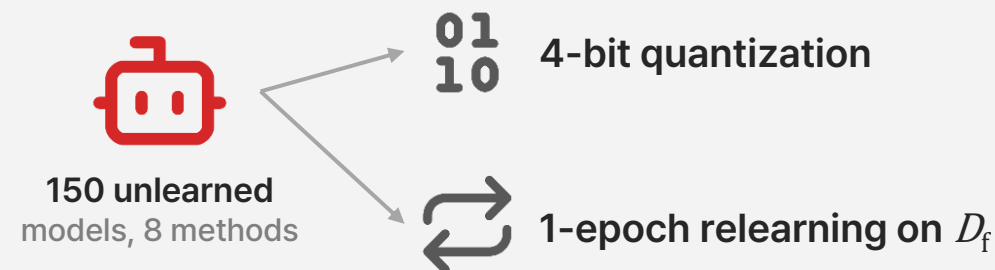
30 trained without D_f



separation → **AUC-ROC**

Robustness

Does the score stay put when the model is perturbed?



score before vs. after → Q, R → harmonic mean
Symmetric (ours): penalize drops and jumps alike

20 metrics

Memorization ×8

MIA ×4

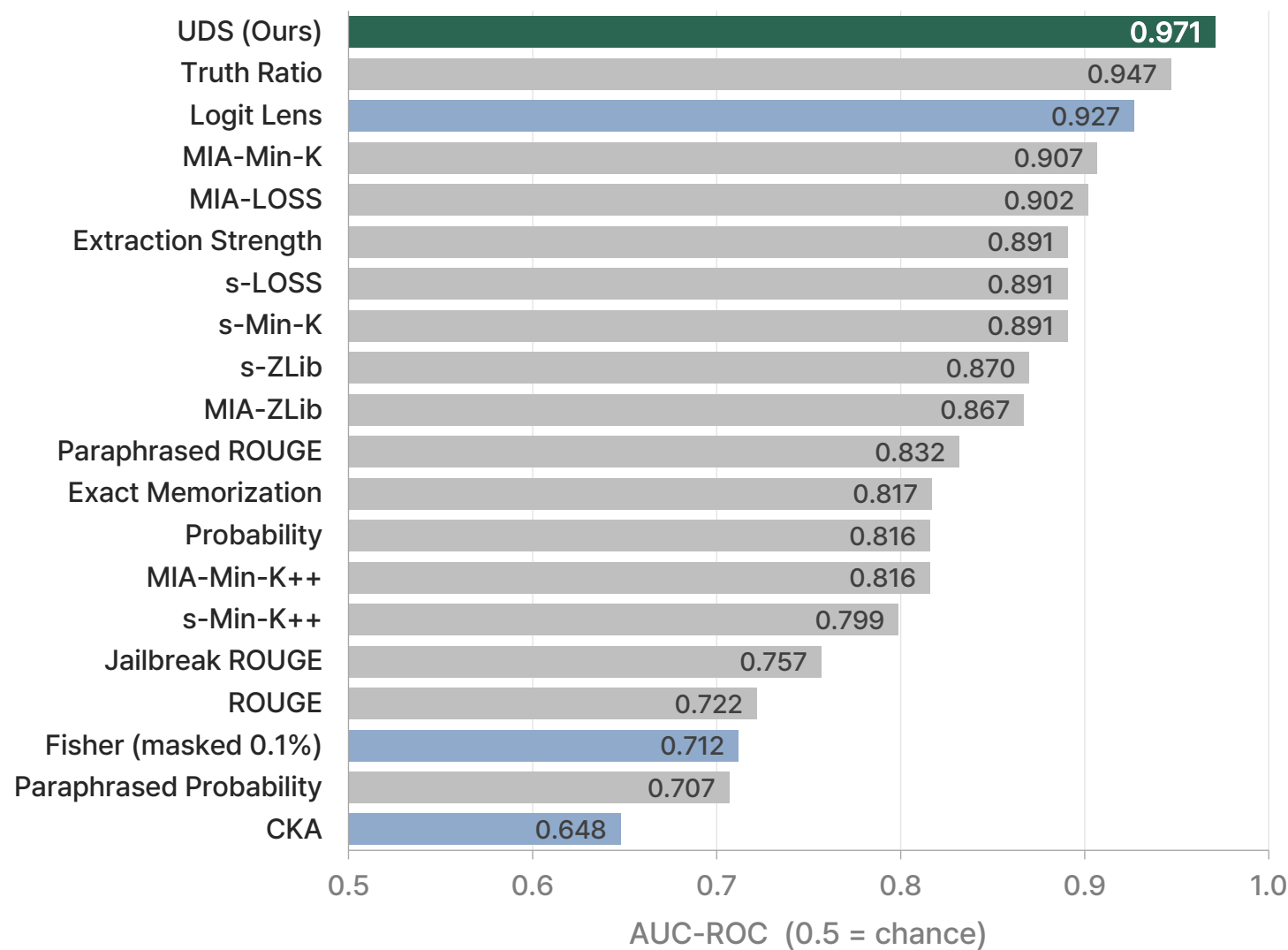
Retain-normalized MIA ×4

White-box: CKA · Logit Lens · Fisher ×3

UDS

Protocol of OpenUnlearning (Dorna et al., 2025) · TOFU forget10 · Llama-3.2-1B-Instruct

Faithfulness: UDS tells the pools apart best



output-level metrics

white-box baselines

UDS (ours)

UDS: AUC 0.971 (1st of 20)

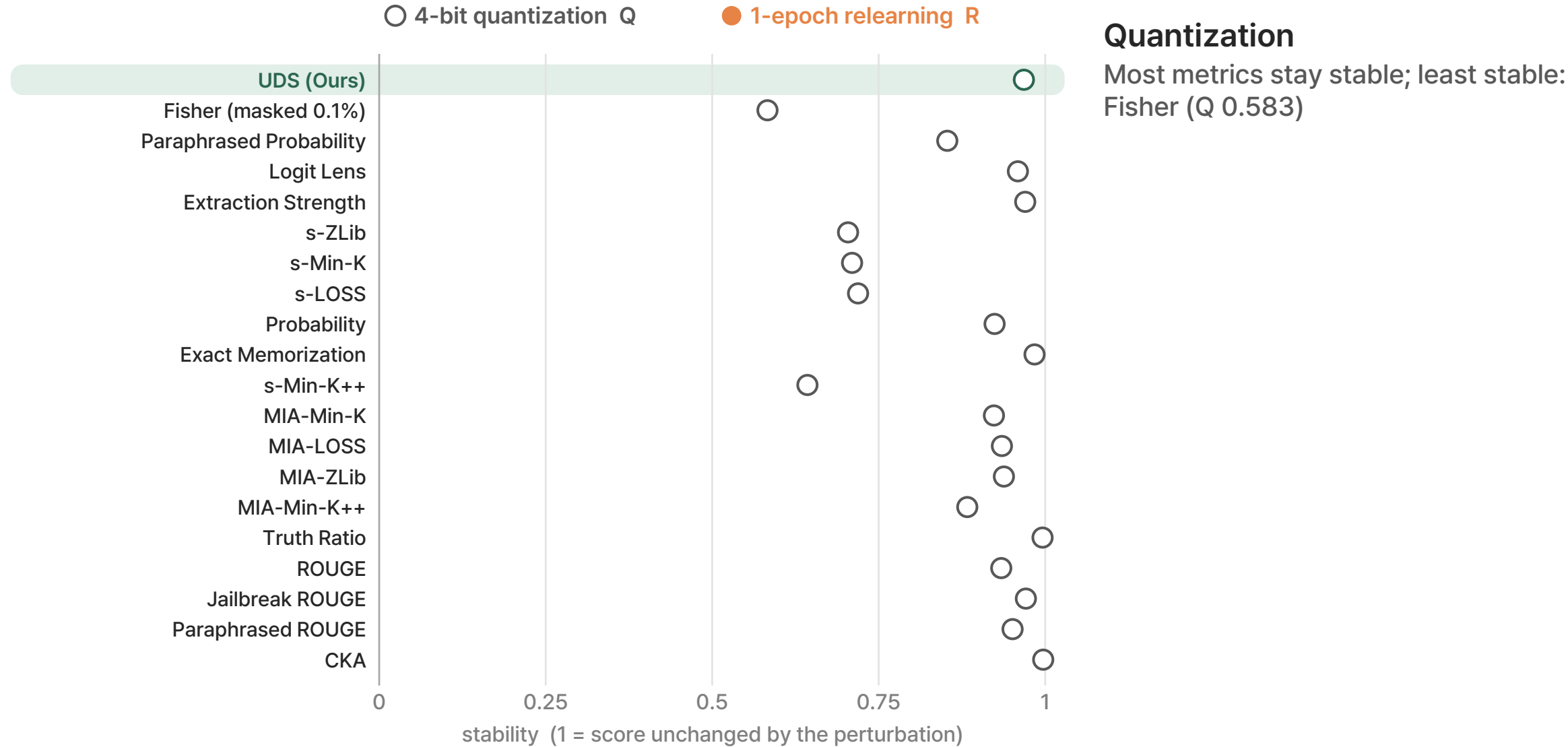
CKA 0.648 · Fisher 0.712

geometry & gradient sensitivity $\neq$ knowledge

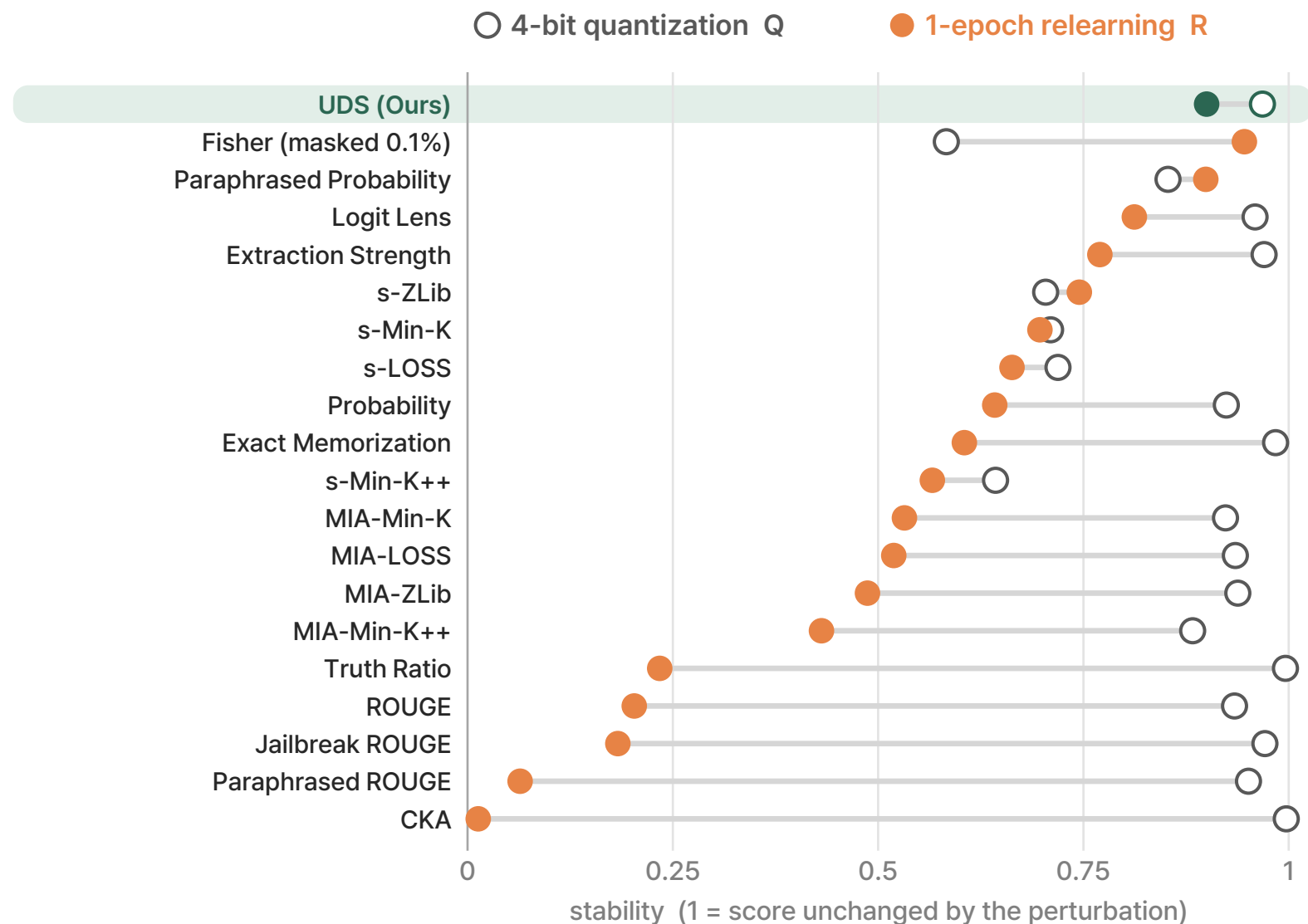
Logit Lens 0.927

reads through a frozen decoder; UDS improves on it with causal intervention

Robustness: **relearning** breaks most metrics



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Quantization

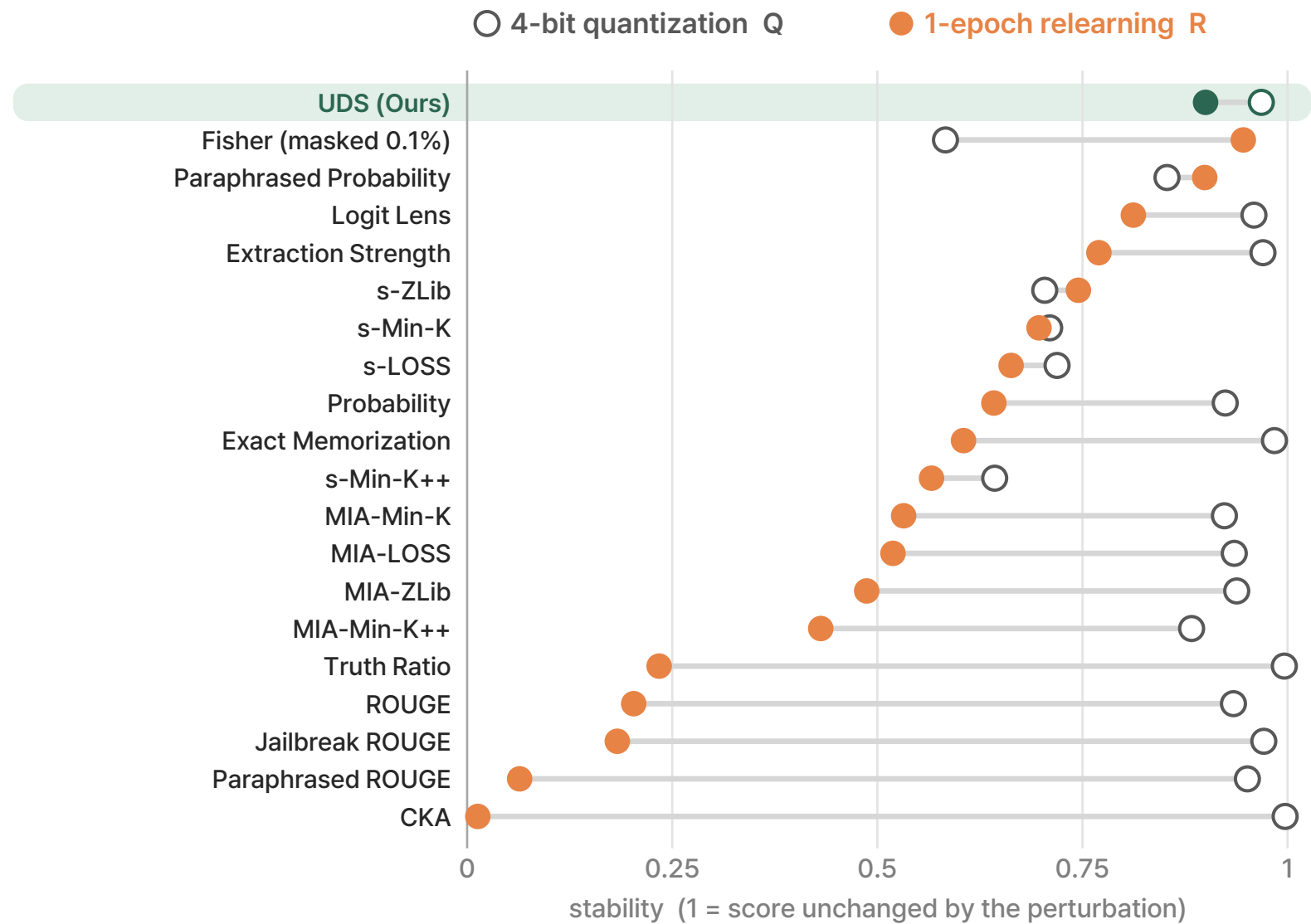
Most metrics stay stable; least stable: Fisher (Q 0.583)

Relearning

Residual knowledge comes back fast:

- ROUGE variants R 0.06–0.20
- Truth Ratio R 0.234
- CKA R 0.013

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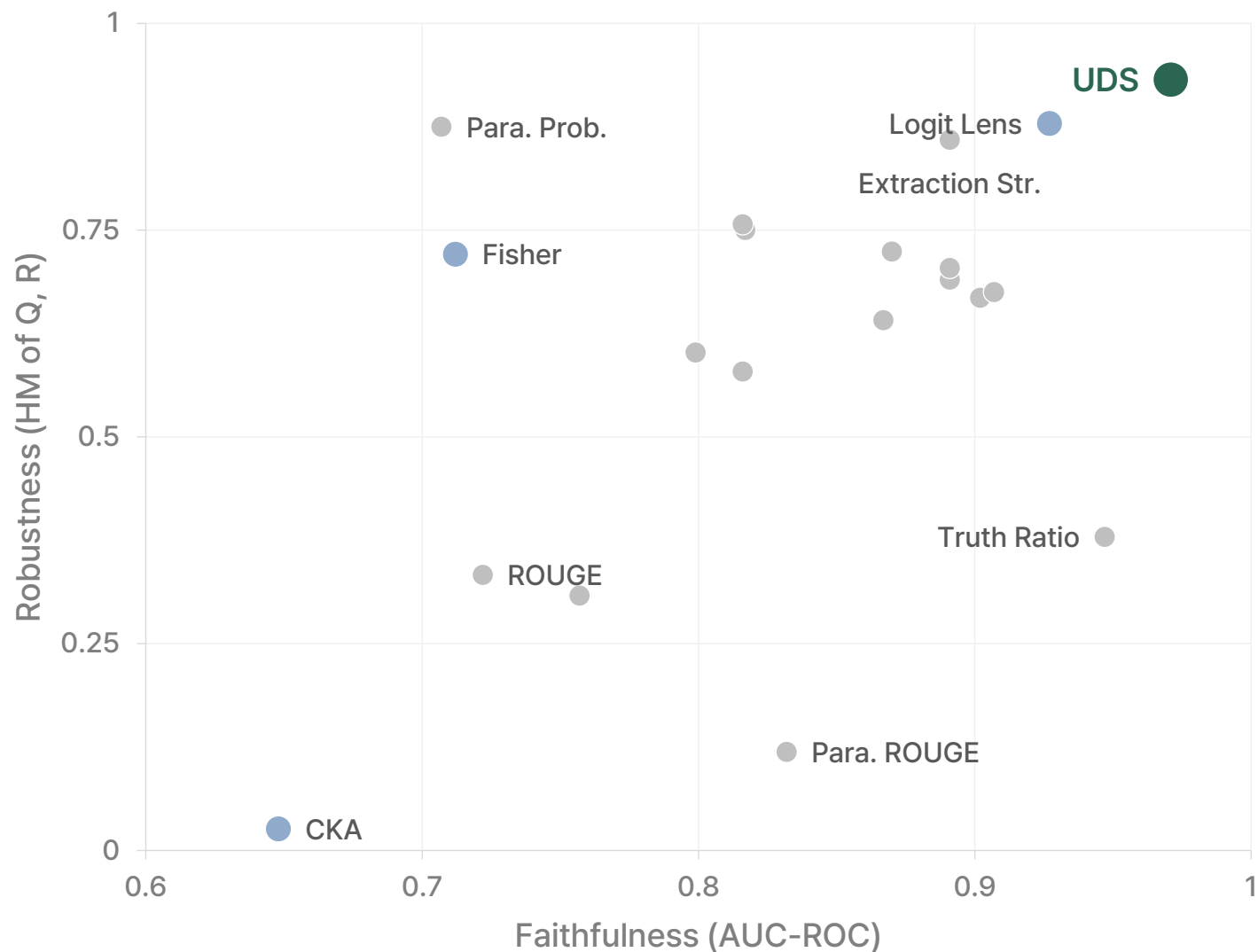
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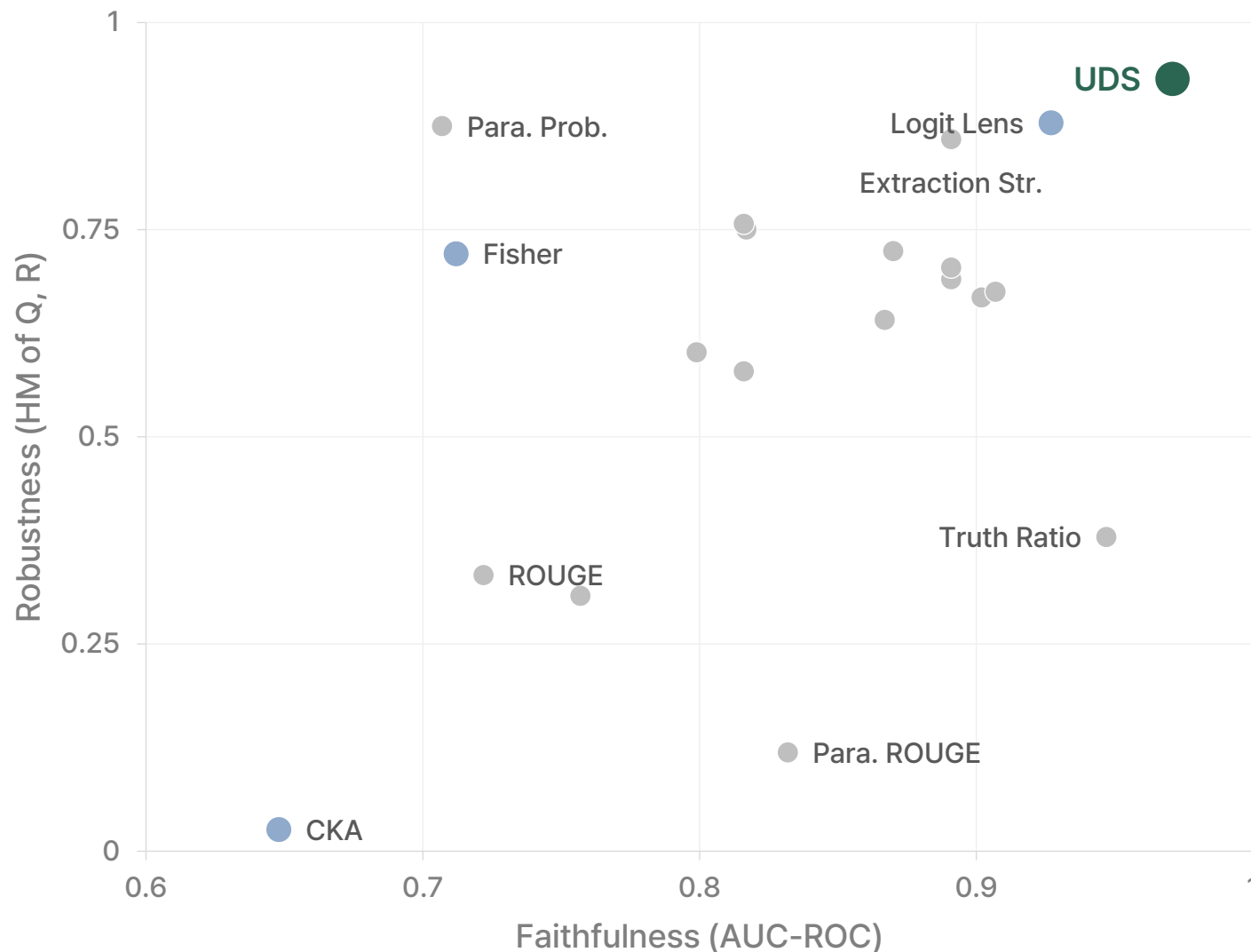
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Faithful **and** robust: UDS sits in the top-right corner



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UDS is 1st on

- faithfulness **0.971**
- robustness **0.932** (Q 0.968 · R 0.900)
- overall **0.951**

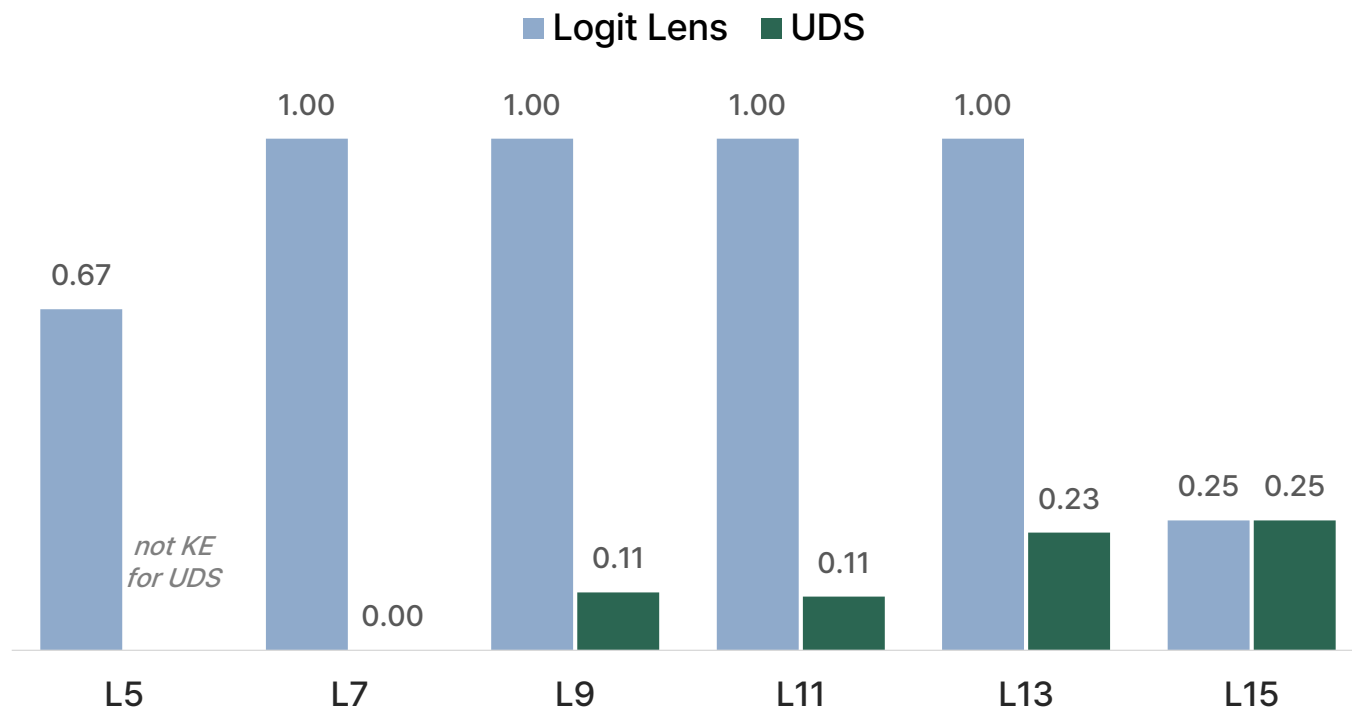
Why?

UDS reads hidden states through the full model's untouched layers, so output-level shifts don't fool it

Case Study – Observational vs. Causal

Logit Lens says “**erased**”, UDS says “**still recoverable**”

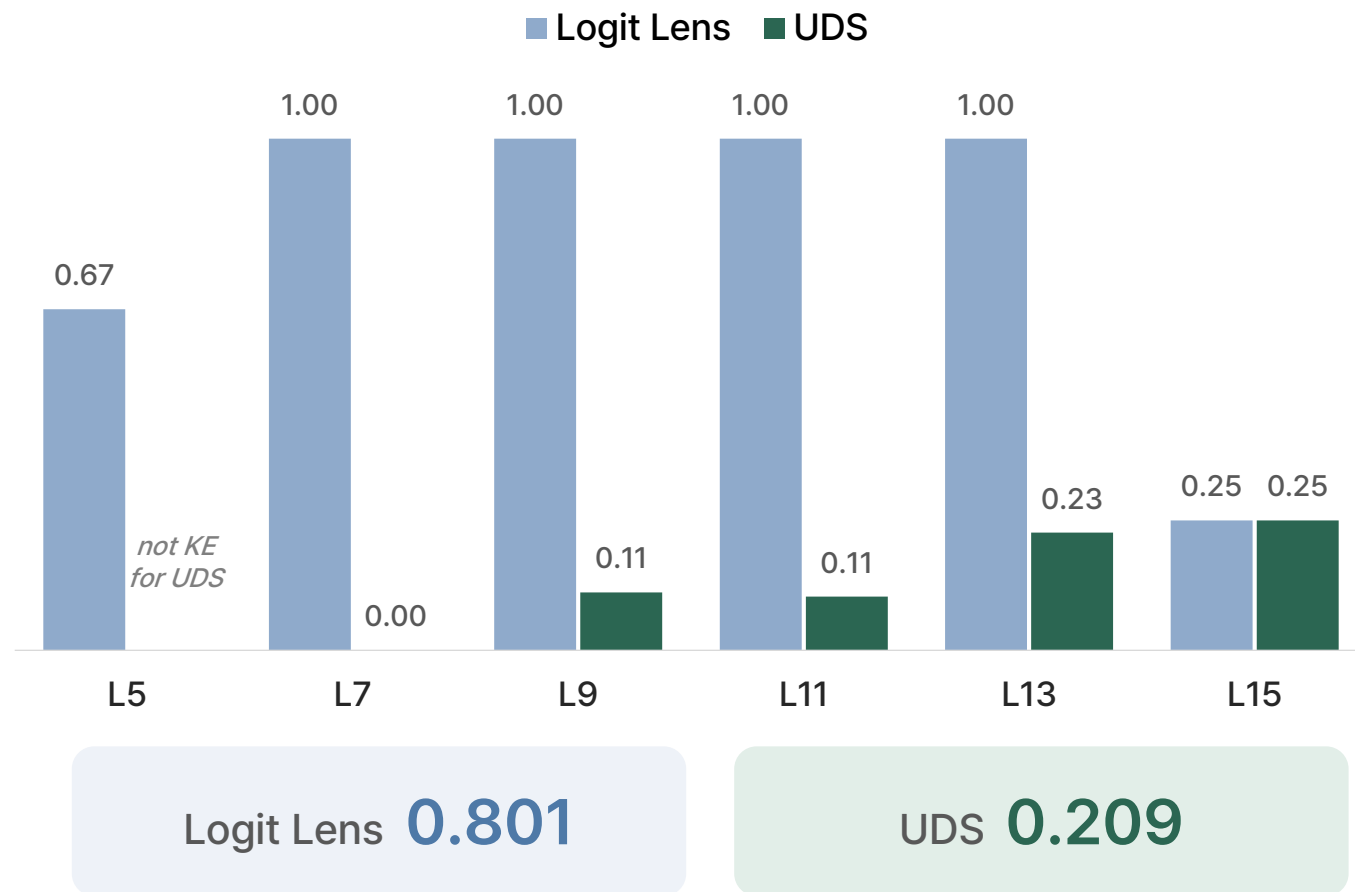
IdkDPO model · entity “**historical fiction**” · Layer Erasure Ratio per layer



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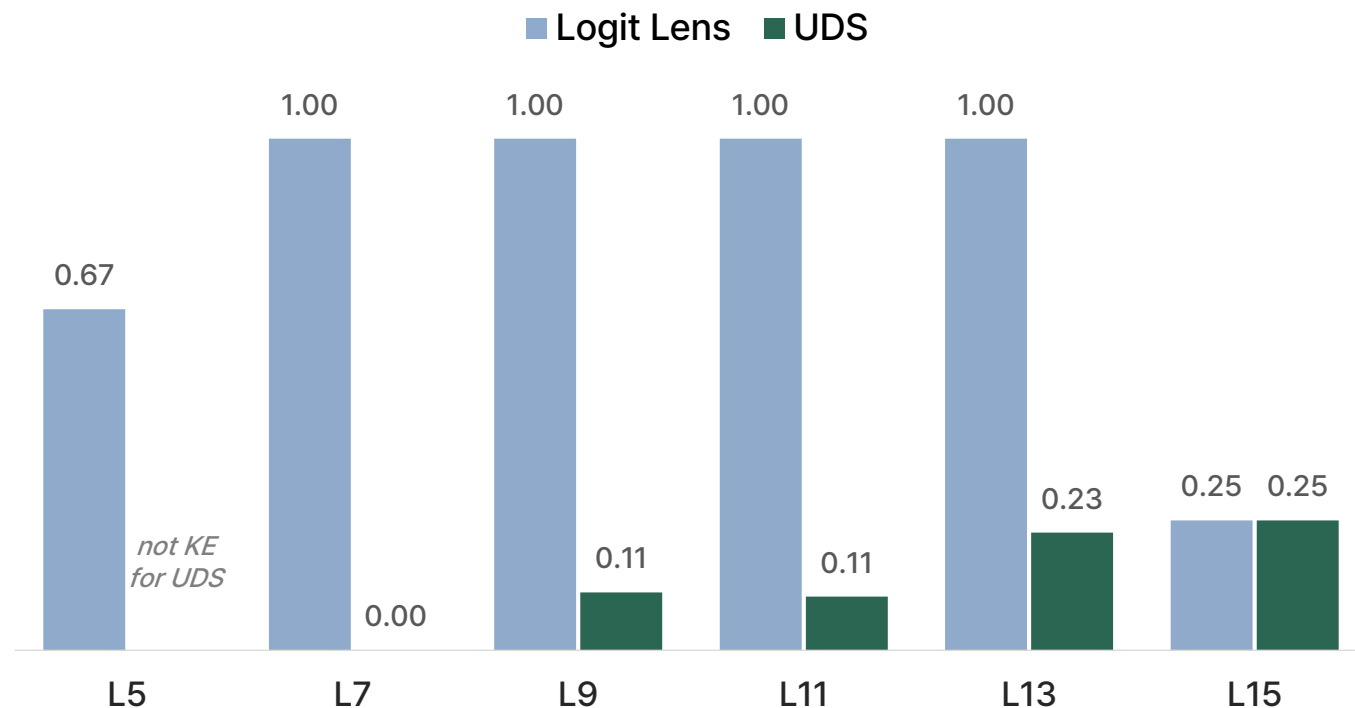
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Logit Lens **0.801**

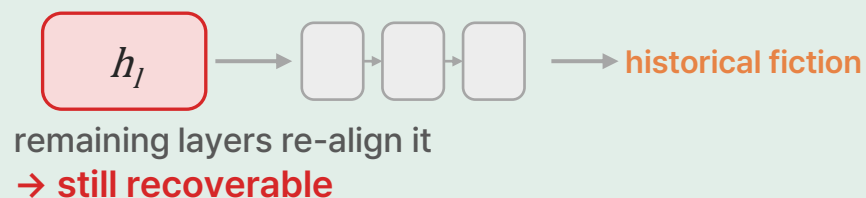
UDS **0.209**

Why they disagree

Logit Lens: frozen unembedding



UDS: patch into M_{full}



Case Study – Heterogeneity of Depth

Same model: **Yes/No** erased, names and titles **untouched**

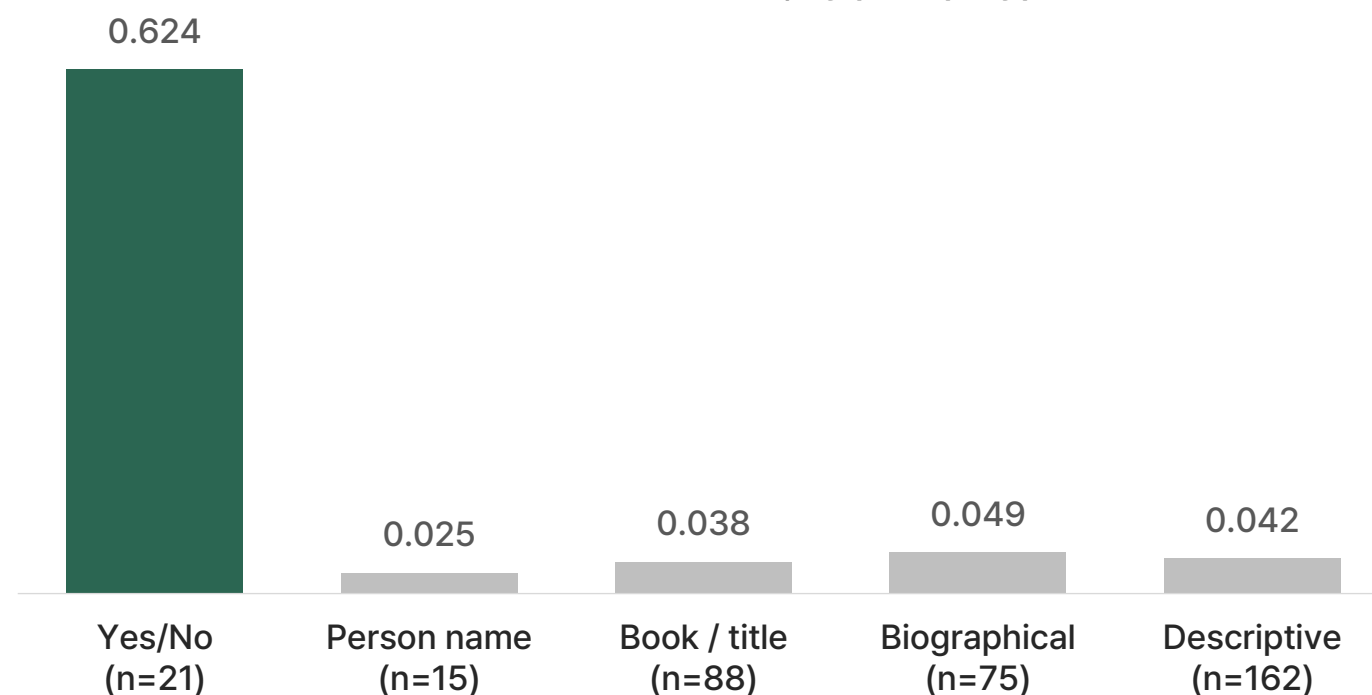
IdkNLL trains the model to answer “I don't know”

Normalized MIA ≈ 0.0 for every IdkNLL config...

...yet UDS separates them: **0.039–0.253**

- **Yes/No:** “I don't know” $\approx$ negating the answer $\rightarrow$ deeper layers change
- **Names, titles, ...:** unrelated to the entity $\rightarrow$ only the output changes

UDS of one IdkNLL model, by prompt type



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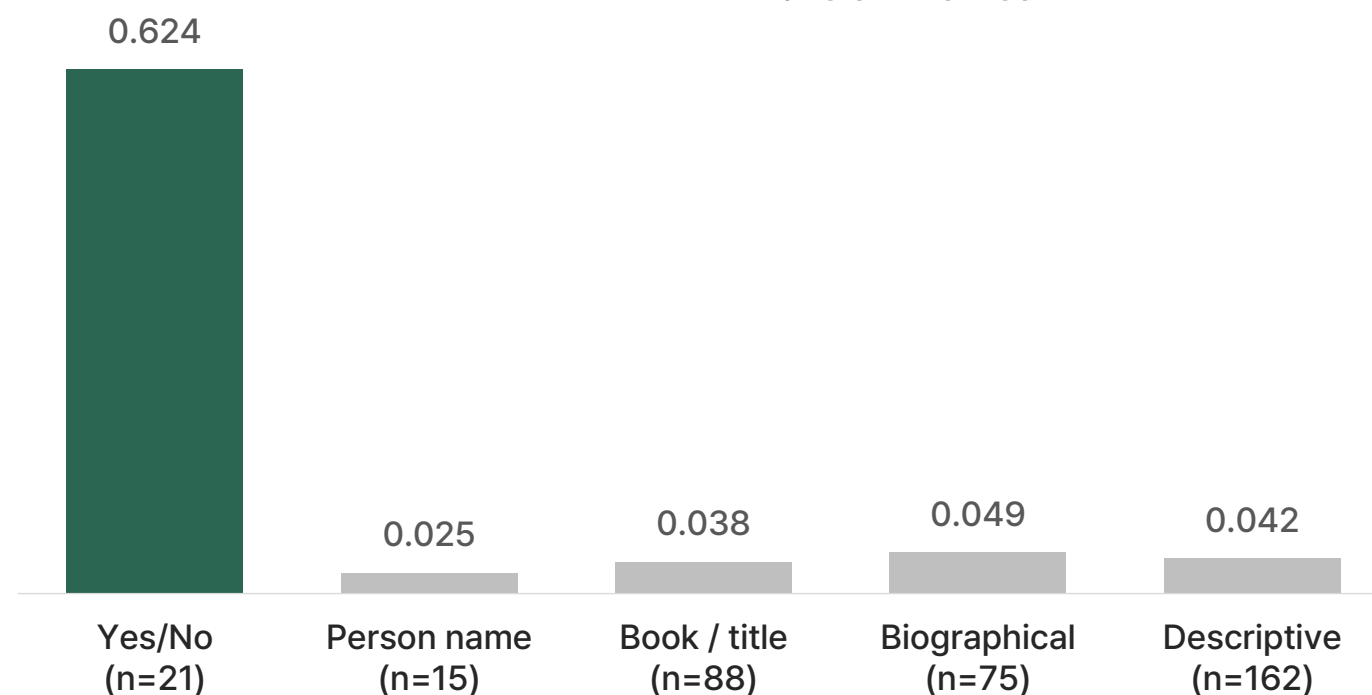
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Per-example UDS: what a method fails to erase

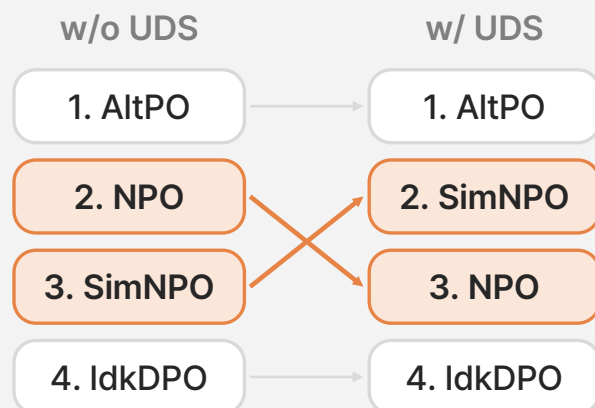
What changes in practice?



Add depth to the privacy axis

$$\text{Privacy} = \text{HM}(\text{MIA}_{\text{agg}}, \text{UDS})$$

method rank (best config)



NPO: MIA 0.875 but UDS 0.619

SimNPO: UDS 0.739 → moves up

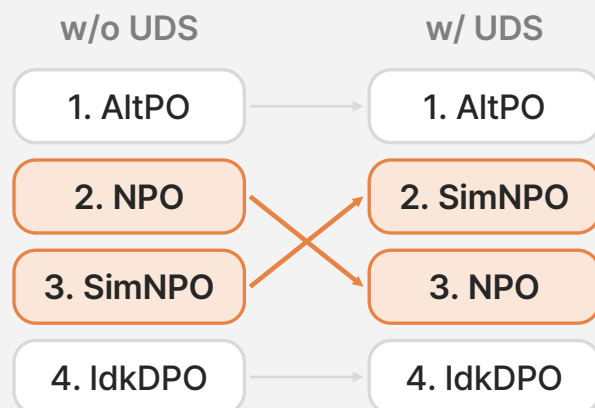
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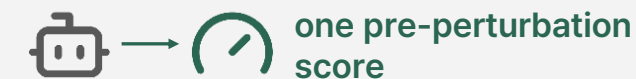
Skip costly re-evaluation

UDS barely moves under quantization (Q 0.968) and relearning (R 0.900)

Today



With UDS



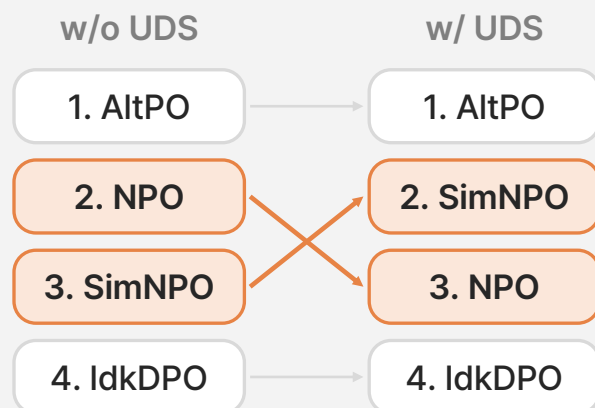
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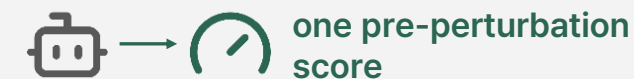
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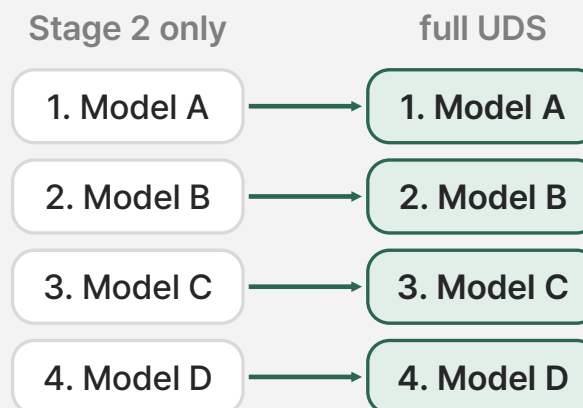


With UDS



No retain model? Run Stage 2 only

Rank models by the Stage 2 drop alone: the order barely changes



$\rho = 0.994$ (150 models)
Stage 1 adds the 0–1 scale

Takeaways

-  **UDS** measures how deeply knowledge is erased: can the full model still decode it from the unlearned model's hidden states?
-  Training-free, causal, dataset-invariant · 0 (intact) → 1 (erased)
-  Most faithful & robust of 20 metrics on 150 unlearned models
-  Reveals residual knowledge that output-level metrics miss



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Project Page

gnueaj.github.io/unlearning-depth-score/

Code & data

github.com/gnueaj/unlearning-depth-score

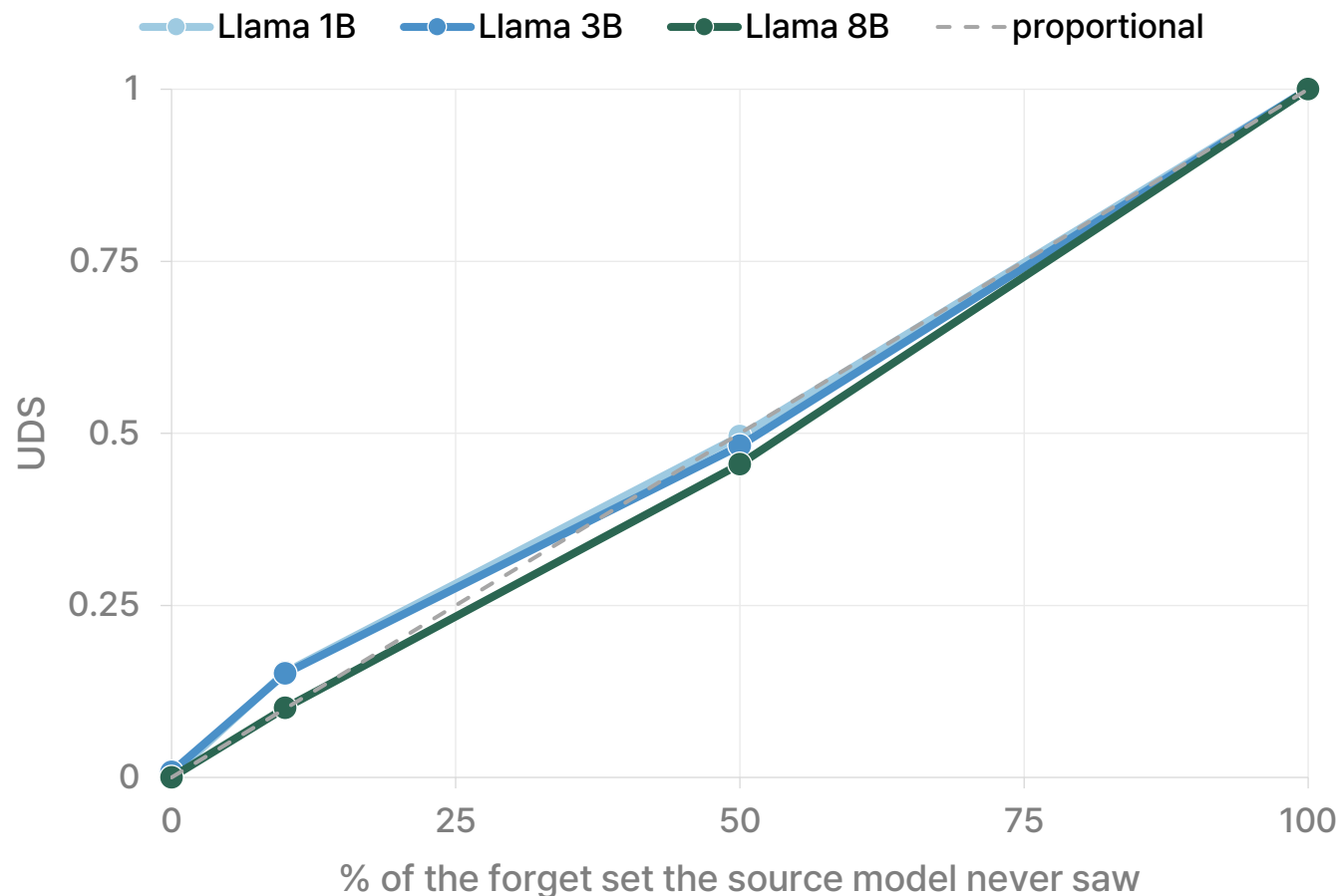


project page

Backup

Extra details for Q&A

Sanity check: UDS tracks what the model never saw



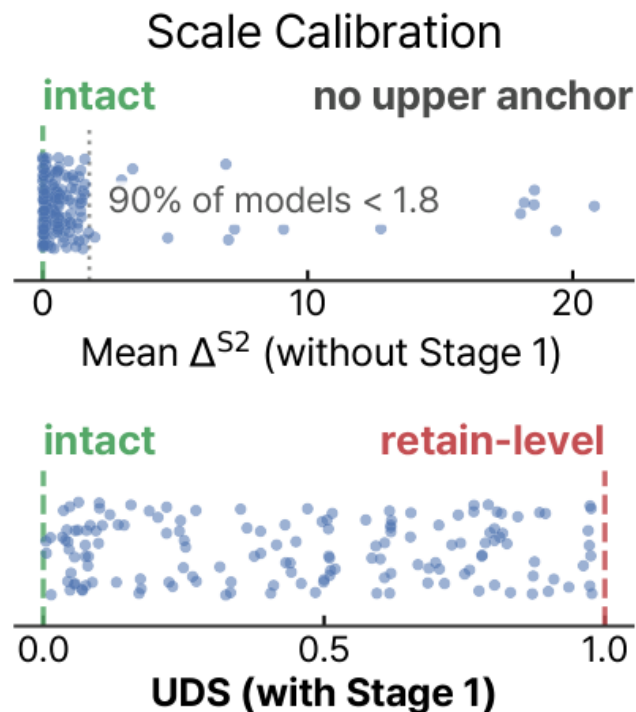
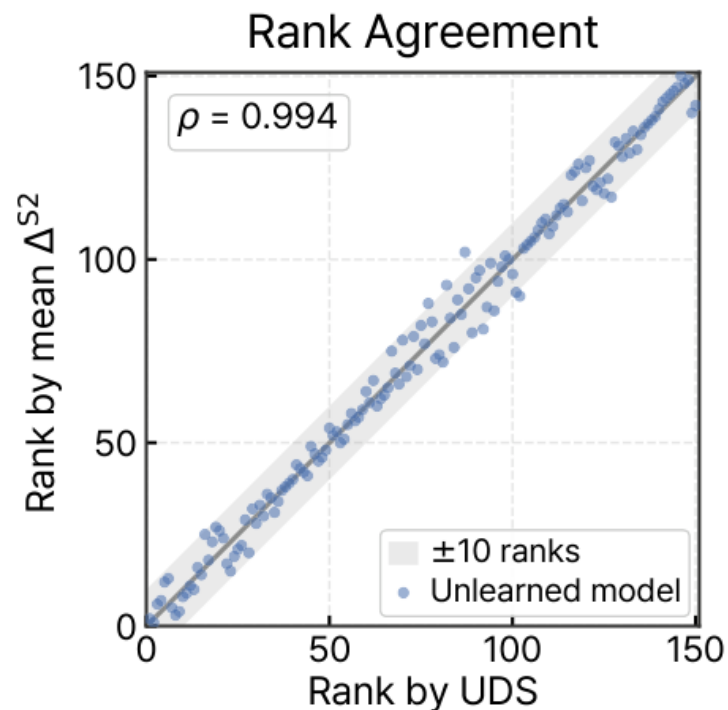
Source models: TOFU retain splits that skipped 0 / 10 / 50 / 100% of the forget set

- **Monotonic & proportional** at 1B, 3B, and 8B
- Slightly lower for bigger models: a small data change perturbs them less

Full meta-evaluation (20 metrics)

Metric	Overall	Faith.	Robust.	Quant. Q	Relearn R
Extraction Strength	0.875	0.891	0.859	0.970	0.770
Exact Memorization	0.782	0.817	0.750	0.984	0.605
Probability	0.786	0.816	0.757	0.924	0.642
Paraphrased Probability	0.782	0.707	0.875	0.853	0.899
Truth Ratio	0.542	0.947	0.379	0.996	0.234
ROUGE	0.456	0.722	0.333	0.934	0.203
Paraphrased ROUGE	0.209	0.832	0.119	0.951	0.064
Jailbreak ROUGE	0.438	0.757	0.308	0.971	0.183
MIA-LOSS	0.767	0.902	0.668	0.935	0.519
MIA-ZLib	0.737	0.867	0.641	0.938	0.487
MIA-Min-K	0.774	0.907	0.675	0.923	0.532
MIA-Min-K++	0.677	0.816	0.579	0.883	0.431
s-LOSS	0.778	0.891	0.690	0.719	0.663
s-ZLib	0.790	0.870	0.724	0.704	0.745
s-Min-K	0.786	0.891	0.704	0.710	0.697
s-Min-K++	0.686	0.799	0.602	0.643	0.566
CKA	0.051	0.648	0.026	0.997	0.013
Fisher (masked 0.1%)	0.716	0.712	0.721	0.583	0.946
Logit Lens	0.902	0.927	0.879	0.959	0.812
UDS (Ours)	0.951	0.971	0.932	0.968	0.900

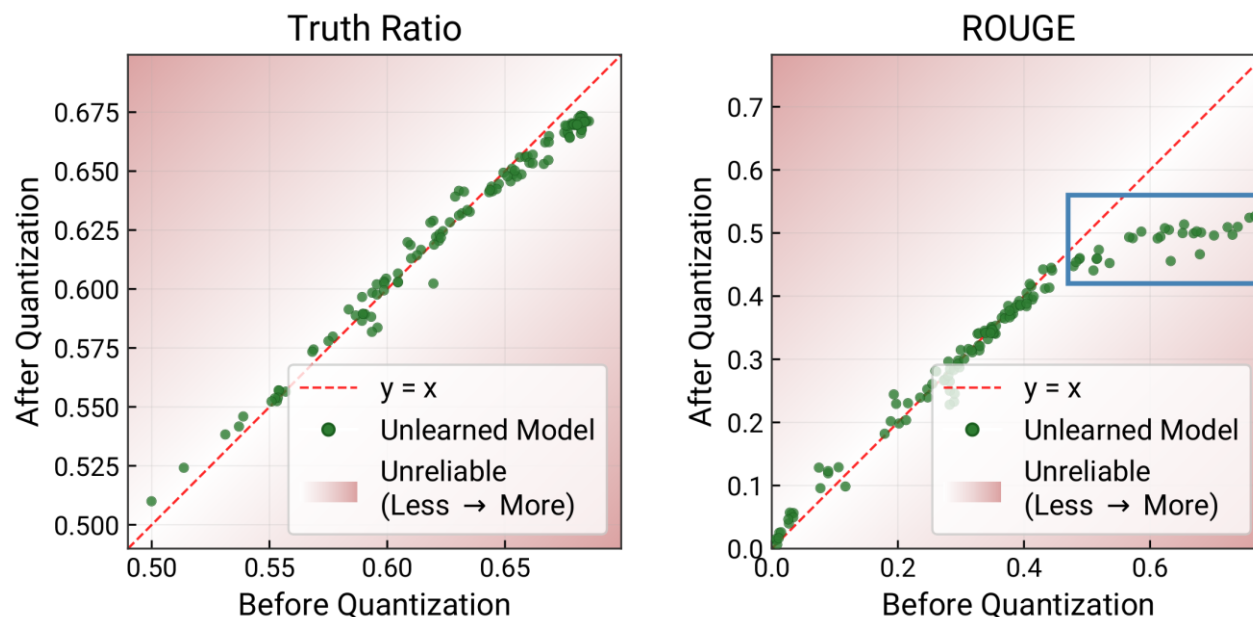
No retain model? Stage 2 alone keeps the ranking



Spearman $\rho = 0.994$

- 96.7% of models within ± 10 ranks
- Without Stage 1 the score has no upper anchor
- Stage 1 calibrates the scale, not the ordering

Why a symmetric robustness score?



$$Q = 1 - \frac{|m' - m|}{|m'| + |m|}$$

$$R = 1 - \frac{|\Delta_{\text{unl}} - \Delta_{\text{ret}}|}{|\Delta_{\text{unl}}| + |\Delta_{\text{ret}}|}$$

Original (OpenUnlearning): $\min(m/m', 1)$ only penalizes recovery

Quantization can break generation → ROUGE drops for reasons unrelated to knowledge, and gets rewarded

Ours: penalize changes in either direction

Runtime per model (seconds)

Metric group	1B	3B	8B
Memorization	28.0	54.7	106.4
Generation	19.7	59.1	160.0
MIA	8.5	16.4	31.4
CKA	2.4	6.6	13.1
Logit Lens	6.9	16.7	26.0
Fisher (masked)	25.8	79.9	211.7
UDS (Ours)	19.1	78.4	184.1

- Stage 1 cached once per forget set
- Teacher forcing: no autoregressive decoding
- Cost grows with #layers (16 / 28 / 32)
- NVIDIA RTX A6000

Limitations

- Needs a retain model for the calibrated scale → Stage 2 alone preserves rankings ($\rho = 0.994$)
- Clipping at 1 hides over-unlearning → monitor UDS together with utility
- Validated on TOFU to match the OpenUnlearning meta-evaluation; MUSE / WMDP are next
- Entity spans + teacher forcing → long-form or multi-step targets remain open